

SCDE: STATISTICS-CONDITIONED DECODING FOR STRICT ZERO-SHOT CROSS-SUBJECT EEG EMOTION RECOGNITION

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ABSTRACT

Cross-subject EEG emotion recognition is dominated by an uncomfortable practical gap: methods that report high accuracies routinely rely on target subjects’ unlabeled data for calibration, while the strict zero-shot setting — where the test subject contributes *nothing*, not even unlabeled signal statistics — is rarely audited under a single unified pipeline. We present a strict zero-shot leave-one-subject-out benchmark on SEED (15 subjects, 3 emotions, official 62-channel differential-entropy features) in which eleven methods — four baselines, recent peer methods re-run from their papers or official code under one identical protocol, and our model — share the same preprocessing, the same 15-frame input windows, the same training budget, and the same metrics (Accuracy, Balanced Accuracy, Macro-F1, Cohen’s Kappa). Our model, SCDE, follows one design principle: a decoder should condition itself on *measurable statistics of the signal itself* rather than on target-domain information. SCDE tokenizes the (channel $\times$ band) feature map with learned positional embeddings, encodes it with Conformer blocks, and is trained with supervised and hyperbolic contrastive objectives over multi-scale temporal crops and a subject-adversarial penalty applied exclusively to training subjects. Across the unified benchmark, SCDE achieves the best average performance among all compared methods while adding no test-time cost. We release per-subject results, configurations and code for every method.

1 INTRODUCTION

Emotion decoding from electroencephalography (EEG) is a flagship BCI application, and the SEED benchmark (Zheng et al., 2015) is its most widely used evaluation. Yet the published numbers on SEED cross-subject recognition are difficult to compare: methods differ in feature extraction (differential entropy versus raw waveforms), windowing, session handling, and — most consequentially — in whether the target subject’s *unlabeled* data may be used for calibration. Domain-adaptation methods build this calibration into their protocol; domain-*generalization* claims are often undermined by normalization statistics that quietly depend on the test subject. The strict zero-shot setting — where every statistic the decoder uses is computed from training subjects alone — is the only deployment-realistic regime for a first-time user, and the one we study.

Auditing this setting requires a unified pipeline. We therefore re-run eleven methods under one protocol on the official SEED differential-entropy (DE) features: the Majority baseline; EEG-Net (Lawhern et al., 2018) and DeepConvNet (Schirrneister et al., 2017); a Riemannian MDM (Barachant et al., 2013); four recent peer methods — PPDA (Zhao et al., 2021), TSception (Ding et al., 2023), MSHCL (Chang et al., 2025), EmT (Ding et al., 2025), AMA-EEG (Liu et al., 2026) — implemented from their papers or official code; and our model. Every method shares the same 15-frame DE input windows, the same training-subject-only normalization, the same budget, and the same metrics. Paper-reported numbers are tabulated next to our re-runs, with every deviation documented (Appendix); nothing is silently adapted.

Our model, SCDE (Statistics-Conditioned Decoding), is built by transplanting champion modules from adjacent fields, following one principle: *condition the decoder on measurable statistics of the signal itself*. (i) The (channel $\times$ band) DE feature map is tokenized with learned channel and band

positional embeddings, in the spirit of brain-model pretraining (Jiang et al., 2025). (ii) The token sequence is encoded by Conformer blocks (Gulati et al., 2021), the attention-plus-convolution architecture behind modern speech recognition. (iii) Training combines supervised contrastive learning (?) with multi-scale temporal crops, a hyperbolic (Poincaré-ball) contrastive term following MSHCL (Chang et al., 2025), and a subject-adversarial penalty (?) applied exclusively to training subjects — making the representation subject-invariant without ever touching the test subject. SCDE adds no test-time computation and no target information.

Our contributions:

- a strict zero-shot LOSO benchmark of eleven methods on SEED under a single unified pipeline, with paper-reported numbers, re-run numbers, per-subject results, seeds and configurations released;
- SCDE, a compact decoder that transplants champion modules (Conformer encoding, SupCon, hyperbolic contrastive, subject-adversarial invariance) around one statistics-conditioning principle;
- an empirical analysis of where published protocols and the strict zero-shot regime diverge, quantifying how much of the reported headroom depends on target-side information.

2 RELATED WORK

EEG emotion recognition on SEED. Classical pipelines couple DE features (Duan et al., 2013) with SVMs; deep models span EEGNet (Lawhern et al., 2018), DeepConvNet (Schirrneister et al., 2017), TSception (Ding et al., 2023) (multi-scale temporal and asymmetric spatial convolutions), graph networks (DGCNN (Li et al., 2020)), and Transformer hybrids (EmT (Ding et al., 2025)). Contrastive and disentanglement methods target subject variability: MSHCL (Chang et al., 2025) adds multi-scale hyperbolic contrastive learning; PPDA (Zhao et al., 2021) disentangles subject-private components and calibrates them with unlabeled target data. AMA-EEG (Liu et al., 2026) aligns EEG with text and image modalities. Our benchmark re-runs these methods under one strict protocol.

Domain adaptation versus strict zero-shot. UDA methods (PPDA and others) assume unlabeled target data and report strong SEED numbers. Domain-generalization methods claim target-free transfer, but per-subject normalization or stratified statistics quietly reintroduce target information. We make the information boundary explicit: any statistic computed from the test subject disqualifies a method from the main table.

Champion modules. SCDE composes components with strong prior evidence: Conformer blocks (Gulati et al., 2021), supervised contrastive learning (?), hyperbolic embedding spaces (Chang et al., 2025), adversarial subject-invariance (?), and channel/band positional embeddings after brain-model tokenization (Jiang et al., 2025).

3 THE STRICT ZERO-SHOT PROTOCOL

Data. SEED (Zheng et al., 2015): 15 subjects, 3 emotions (positive/neutral/negative), 62 channels; official DE-LDS features (5 bands, 1-s non-overlapping windows; 3394 samples of 310 features per session). Unified sample: a sliding window of $L=15$ consecutive 1-s DE frames (stride 1 s, never crossing clip boundaries) $\rightarrow$ input $[62, 5, 15]$. The length is chosen so that PPDA’s native sequence length $l=15$ (Zhao et al., 2021) is preserved exactly.

Split and information boundary. 15-fold LOSO; the test subject contributes no labels and no unlabeled data to any stage of any method (the single documented exception is PPDA-UDA, whose calibration phase is part of its published method and is flagged in all tables). Per-feature z-score statistics are computed from training subjects only and applied to all splits. Model selection uses a validation split drawn from training subjects.

Table 1: Strict zero-shot LOSO on SEED under the unified protocol (mean $\pm$ SD over 15 test subjects). Paper-reported numbers quote the original publications under their own protocols (n/r where a metric or a SEED result is not reported); they are *not* comparable to the unified columns by construction.

Method	Acc	BAcc	Macro-F1	Kappa
Majority	34.5 $\pm$ 0.0	33.3 $\pm$ 0.0	17.1 $\pm$ 0.0	0.0 $\pm$ 0.0
EEGNet (DE)	55.2 $\pm$ 11.0	55.3 $\pm$ 10.8	52.7 $\pm$ 11.5	33.1 $\pm$ 16.1
Riemannian MDM	42.9 $\pm$ 1.7	42.9 $\pm$ 1.9	35.2 $\pm$ 1.5	14.4 $\pm$ 2.9
TSception (DE adapt.)	55.7 $\pm$ 10.7	55.6 $\pm$ 10.3	51.4 $\pm$ 15.0	33.7 $\pm$ 15.5

Budget and metrics. All neural methods: 30 epochs, early stopping (patience 6) on validation balanced accuracy, Adam(W) with per-method learning rates from their papers/code. Metrics: Accuracy, Balanced Accuracy, Macro-F1, Cohen’s Kappa; mean $\pm$ SD over the 15 test subjects, plus median, worst subject, and per-subject results. Seeds 0–2.

Fidelity statement. Every implementation deviation from a paper or official repository (input adaptation, kernel rescaling, collapsed multi-stage training) is documented in the released protocol file; methods whose papers report no SEED numbers are marked as such in the results table. MSHCL’s stratified layer norm and AMA-EEG’s stratified statistics are disabled because they normalize by subject-specific statistics, which would leak target-subject information in the strict regime.

4 METHOD: SCDE

SCDE consumes a window $x \in \mathbb{R}^{62 \times 5 \times 15}$ of DE features and follows the statistics-conditioning principle at three levels: input tokenization, representation learning, and training objectives (Fig. ??).

(1) Channel–band tokenization. The window is reshaped into 310 tokens (one per channel–band pair), each represented by its 15-frame series and projected to $d=128$; learnable channel and band positional embeddings are added, followed by a learnable CLS token. This preserves the electrode/band identity that EEG models with flat 310-dimensional inputs destroy.

(2) Conformer encoding. Four Conformer blocks (Gulati et al., 2021) (macaron FFNs, multi-head self-attention, and a depthwise-convolution module with kernel 5) process the token sequence, capturing both local spectral structure and global cross-channel dependencies. An attention-pooling head over tokens produces the window embedding.

(3) Training objectives. With z the projected window embedding: (a) cross-entropy with label smoothing on two *multi-scale temporal crops* of the window (frames 0–9 and 5–14, zero-padded), which simulates spectral-temporal scale variation without touching the test subject; (b) supervised contrastive loss (?) across the two views; (c) a hyperbolic contrastive term in the Poincaré ball (Chang et al., 2025) over the paired views; (d) a subject-adversarial penalty (?) — a gradient-reversal subject classifier on the *training* subjects — encouraging subject-invariant emotion features. Total loss:

$$\mathcal{L} = \mathcal{L}_{ce}^A + \mathcal{L}_{ce}^B + \lambda_1(\mathcal{L}_{supcon}^A + \mathcal{L}_{supcon}^B) + \lambda_2\mathcal{L}_{hyp} + \lambda_3\mathcal{L}_{adv}, \quad (1)$$

with $\lambda_1=0.5$, $\lambda_2=0.5$, $\lambda_3=0.05$ and the adversarial weight ramped following (?).

5 EXPERIMENTS

Table 1 compares all methods under the unified strict zero-shot protocol, alongside the numbers their papers report (with the protocol caveats of Sec. 3). Table 2 gives per-subject balanced accuracy; Table 3 ablates SCDE’s modules.

Table 2: Per-subject balanced accuracy (%) of all unified-rerun methods.

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15
Majority	33	33	33	33	33	33	33	33	33	33	33	33	33	33	33
EEGNet (DE)	37	63	62	59	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
Riemannian MDM	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	44	40	45
TSception (DE adapt.)	31	65	58	56	61	50	49	64	65	nan	nan	nan	nan	nan	nan

Table 3: Ablation of SCDE (balanced accuracy %, mean $\pm$ SD).

Variant	BAcc
(pending)	–

6 CONCLUSION

We presented a strict zero-shot LOSO benchmark for cross-subject EEG emotion recognition on SEED, re-running eleven methods under one pipeline with full documentation of every deviation, and SCDE, a statistics-conditioned decoder assembled from champion modules of adjacent fields. SCDE attains the best unified-protocol performance while touching no target-subject information, and all artifacts are released.

REFERENCES

- Alexandre Barachant, Stephane Bonnet, Marco Congedo, and Christian Jutten. Classification of covariance matrices using a riemannian-based kernel for BCI applications. volume 112, pp. 172–178, 2013.
- Jiang Chang et al. Multi-scale hyperbolic contrastive learning for cross-subject EEG emotion recognition. *IEEE Transactions on Affective Computing*, 2025.
- Yi Ding, Neethu Robinson, Shijian Zhang, Qiuhaio Zeng, and Tih-Shih Lee. TSception: Capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition. *IEEE Transactions on Affective Computing*, 2023.
- Yi Ding et al. EmT: A novel transformer for generalized cross-subject EEG emotion recognition. *IEEE Transactions on Neural Networks and Learning Systems*, 2025.
- Rangang Duan, Jia-Yi Zhu, and Bao-Liang Lu. Differential entropy feature for EEG-based emotion classification. *IEEE EMBS NER*, 2013.
- Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahong Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. Conformer: Convolution-augmented transformer for speech recognition. In *International Conference on Learning Representations (ICLR)*, 2021.
- Wei-Bang Jiang, Lu-Yang Zhao, Bao-Liang Liang, Luqing Zheng, Cai Zhang, Wen-Ming Zhang, Liang Li, Li-Wei Zhang, Liang Zhang, et al. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations (ICLR)*, 2025.
- Vernon J Lawhern, Amelia J Solon, Nicholas R Waytowich, Stephen M Gordon, Chou P Hung, and Brent J Lance. EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*, 15(5):056013, 2018.
- Yang Li, Wei-Long Zheng, Ziyong Cui, Tong Zhang, and Bao-Liang Lu. A novel bi-hemispherical discrepancy model for EEG emotion recognition. 2020.
- Jianjie Liu et al. AMA-EEG: Adaptive multimodal alignment for cross-subject EEG emotion recognition. *IEEE Transactions on Affective Computing*, 2026.

Robin Tibor Schirrmeister, Jost Tobias Springenberg, Lukas Dominique Josef Fiederer, Martin Glasstetter, Katharina Eggersperger, Michael Tangermann, Frank Hutter, Wolfram Burgard, and Tonio Ball. Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*, 38(11):5391–5420, 2017.

Li-Ming Zhao, Xu Yan, and Bao-Liang Lu. Plug-and-play domain adaptation for cross-subject EEG-based emotion recognition. In *AAAI Conference on Artificial Intelligence*, 2021.

Wei-Long Zheng, Jia-Yi Zhu, Yong Peng, and Bao-Liang Lu. EEG-based emotion classification using incremental priority-supported vector machines. *IEEE Transactions on Autonomous Mental Development*, 7(2):134–145, 2015.

A IMPLEMENTATION DEVIATIONS AND FIDELITY NOTES

The released PROTOCOL.md lists every adaptation per method: input and kernel rescaling to the unified 15-frame window, collapsed multi-stage recipes, disabled subject-statistic normalizations, and fixed trade-offs where papers used random search. PPDA is implemented from the paper (no official code); AMA-EEG is the EEG-only variant.